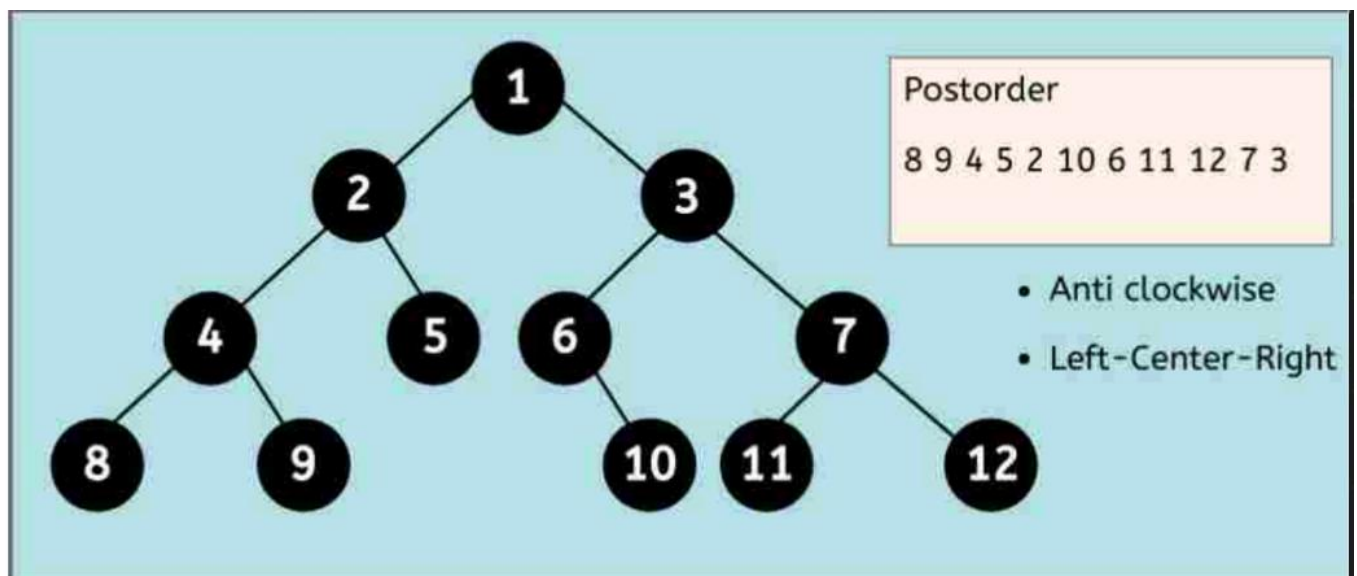


TREE-01

Q4)

Question:

DS	Session	Tree 1	Question Information	Level 1	Challenge 64
Question description Darsh, Ratik, Swathy are good friends, They are studying Pre-final year B.E. in reputed institution. Swathy's uncle was a DS teacher in school of computing technologies. He asks to make an application for identifying postorder Traversal of the Tree You have to do the work for the development of the code of their thinking. Functional Description					



Constraints

$T \leq 100$

$1 \leq n \leq 1000$

$0 \leq k < n$

PROGRAM:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node *left, *right;
7  };
8
9  struct Node* create(int x)
10 {
11     struct Node *n = malloc(sizeof(struct Node));
12     n->data = x;
13     n->left = n->right = NULL;
14     return n;
15 }
16
17 struct Node* insert(struct Node *root, int x)
18 {
19     if (root == NULL)
20         return create(x);
21
22     if (x < root->data)
23         root->left = insert(root->left, x);
24     else
25         root->right = insert(root->right, x);
26
27     return root;
28 }
29
30 void postorder(struct Node *root)
31 {
32     if (root == NULL)
33         return;
34
35     postorder(root->left);
36     postorder(root->right);
37     printf("%d ", root->data);
38 }
39
40 int main()
41 {
42     int t, n, i, x;
43     struct Node *root;
44
45     scanf("%d", &t);
46
47     while (t--)
48     {
49         scanf("%d", &n);
50         root = NULL;
51
52         for (i = 0; i < n; i++)
53         {
54             scanf("%d", &x);
55             root = insert(root, x);
56         }
57
58         postorder(root);
59         printf("\n");
60     }
61
62     return 0;
63 }
```

OUTPUT:

TEST CASE 1:

```
1
7
90 70 50 75 130 110 150
50 75 70 110 150 130 90
```

TEST CASE 2:

```
1
9
50 23 56 89 43 75 130 110 150
43 23 75 110 150 130 89 56 50
```